Ruger Mini-14 & Mini Thirty

Ruger • Garand-Style Rotating Bolt Carbines (5.56x45mm / .223 Rem & 7.62x39mm)

Manufacturer:	Ruger	**Category:**	Rifles & Carbines
Action Type:	Short-Stroke Gas Piston Semi-Automatic	**Era / Service:**	Modern Classic Carbine (1973–Present)
Calibers:	5.56x45mm NATO, .223 Remington, 7.62x39mm, 300 AAC Blackout	**Covered Models:**	Mini-14 Ranch Rifle, Mini-14 Tactical, Mini Thirty (7.62x39mm), AC-556 (Selective Fire)

KEY ARMORER TASKS & BENCH OPERATIONS

Core Procedures: Gas Block Screw Gap Equalization (30 in-lbs) • Operating Rod Slide Channel Polishing • Garand-Style Trigger Group Service • Gas Bushing Diameter Tuning • Magazine Latch Fitting

CRITICAL ARMORER TOLERANCES & SPECIFICATIONS

SPECIFICATION / CLEARANCE	ARMORER TOLERANCE & BENCH SPECIFICATION
GAS BLOCK SCREWS	Torque to exactly 30 in-lbs; armorer MUST measure gap between gas block halves with feeler gauge to ensure equal spacing on all 4 corners
GAS BUSHING	Internal gas orifice bushing under gas block controls gas volume (0.080" factory standard; reduced to 0.045"–0.050" for taming violent brass ejection)
OPERATING SYSTEM	Garand-derived operating rod cycling an exposed rotating two-lug bolt
TRIGGER TAKEDOWN	Unlatching rear of trigger guard allows complete trigger group to drop out bottom of stock

IMMEDIATE ACTION & MALFUNCTION DIAGNOSTIC MATRIX

MALFUNCTION CONDITION	PROBABLE CAUSES	ARMORER BENCH REMEDY
Stoppage 1: Violent Over-Gassed Ejection (Brass Thrown 30+ Feet) Pistol throws spent brass 30 to 45 feet into next shooting bay; op-rod batters receiver face	• Factory gas bushing orifice intentionally drilled large for reliable cycling with weak ammunition. • Excessive gas pressure slamming op-rod rearward.	**Gunsmith Action: Remove gas block screws. Pull out factory 0.080" gas bushing. Install reduced-diameter gas bushing (0.045"–0.050"). Re-torque gas block screws in criss-cross pattern to 30 in-lbs, ensuring equal gap on both sides with feeler gauge.**

CURATED BENCH VIDEO REFERENCES

PROCEDURE TITLE	INSTRUCTOR / CREATOR	TECHNICAL PROCEDURE SUMMARY
Ruger Mini-14 Complete Takedown, Gas Block Tuning & Accuracy	MidwayUSA / Larry Potterfield	Teardown of the Mini-14, equalizing gas block screw gaps, disassembling the bolt with a bolt tool, and tuning the trigger group.